Geometry: Unit 13, Lesson 9

1. Line segment ZM is perpendicular to line segment RL. If the slope of $\overline{ZM}$ is -2, and point R is located at $(8, -3)$, what points could be the location of point L?

2. Line segment KR is perpendicular to line segment LP. If the slope of $\overline{KR}$ is $-\frac{1}{3}$, and point L is located at $(-1, 4)$. Point P is located at $(x, 16)$. What is the value of x?

3. Line segment DW has endpoints at point $D(-5, -4)$ and at point $W(1, -12)$. What is the midpoint of $\overline{DW}$?